

Evaluating Systematic Momentum Strategies Against Passive Index Investing

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GitHub: https://github.com/tejasmanojav/momentum_trading/tree/main

Objective

I have long been fascinated by the mathematical elegance of passive index investing. After reading several of John Bogle's books and studying his interviews, his core thesis made inherent sense to me: minimize frictional costs and allow the natural compounding of the broader market to work uninterrupted. Furthermore, studying historical financial blunders repeatedly showed me how emotionally driven, reactive trading can erode wealth over time. However, while I fundamentally believe in Bogle's passive foundation, I wanted to rigorously test it myself rather than accept it blindly. I set out to determine whether simple, mathematically disciplined momentum rules could meaningfully improve or complement a long-term buy-and-hold approach—specifically by limiting downside risk during major market drawdowns. To find out, I built a custom backtesting engine in Python to evaluate and refine these strategies against 20 years of historical data.

Hypothesis

"Can simple momentum trading strategies outperform a passive buy-and-hold approach for major indices?"

I structured my code to test three main assumptions:

1. Momentum strategies can help reduce large drawdowns during market crashes.
2. The Relative Strength Index (RSI) might perform differently depending on market volatility.
3. Adding complexity and adaptive risk management to the trading rules will improve the consistency of the returns.

Note: I focused on indices as they are not heavily swayed by unexpected news or earnings reports tied to a single company. I used the `yfinance` library to pull 20 years of daily price data across five major diversified ETFs (SPY, QQQ, DIA, IWM, VTI).

Methodology & Strategy Evolution

1. Baseline: Buy & Hold

To properly evaluate my algorithmic strategies, I first established a baseline control group. The Buy & Hold strategy represents purely passive investing: buying the indices on day one and holding them for 20 years, riding out all market crashes and bull runs.

2. Trend Following: Moving Average (MA)

My first test of momentum was a simple trend-following mechanic: the algorithm buys when the price moves above a moving average and sells when it drops below. I evaluated various price windows:

- **Longer baselines (e.g., 200-day MA):** These triggered fewer trades and protected against massive drops, but they were slow to react to market changes.
- **Shorter baselines (e.g., 20-day MA):** These reacted quickly but generated excessive trades when the market was trending flat (moving sideways), resulting in a low win rate and lower overall returns.

3. Noise Reduction: MA + Buffer

To address the frequent entry/exit patterns during flat, sideways markets, I realized I needed a way to ensure the trading signal was actually stable before executing.

- **The Fix:** I added a 2% percentage buffer (**epsilon = 0.02**) so the price had to move beyond that threshold before triggering a trade. I did a coarse "hyperparameter search" but did not find a compelling alternative.
- Using a percentage of the price rather than an absolute dollar amount allowed the logic to work as a scalable hyperparameter across all five indices, successfully reducing the "jitter" trades.

4. Risk Management: MA + Fixed Stop Loss

While testing, I noticed that there were some trades with large losses wiping out cumulative profits. To mathematically cap my downside risk, I programmed a 5% fixed trailing stop-loss.

- **The Trade-off:** Setting the trailing stop-loss too conservatively resulted in early profit-taking. While it protected the portfolio from large drawdowns, it frequently cut winning trades short during normal market volatility, which severely reduced my expected returns.

5. Adaptive Risk Management: MA + ATR Stop Loss

Because a fixed 5% stop-loss choked the algorithm, I realized I needed a dynamic metric. By integrating the Average True Range (ATR), the algorithm can adapt its stop-loss based on the actual daily volatility of the asset, giving it "room to breathe" during volatile periods.

- **The 200-Day MA Winner:** The data showed that an ATR 4.0x multiplier prevented premature exits for long-term strategies. By using a wide, volatility-adjusted buffer, it significantly outperformed the fixed 5% stop (a 383% total return vs. 330%). It successfully ignored market noise while keeping trade churn manageable, averaging roughly 60 trades per ticker (302 total trades across the portfolio) compared to a much higher churn for a tighter 2.0x ATR.

- **The 50-Day MA Paradox:** Interestingly, for the shorter 50-day window, the fixed 5% stop remained superior. This observation suggests that shorter-term strategies may benefit from tighter exit conditions, while longer-term strategies appear to benefit from greater flexibility during periods of volatility.

6. Mean Reversion: RSI Strategy

As an alternative to trend following, I tested a mean-reversion strategy using the 14-day Relative Strength Index (RSI). For selected shorter, highly volatile 5-year windows, RSI seemed to perform better by capitalizing on assets that were mathematically "oversold" (<30) or "overbought" (>70).

7. Synthesis: The Confluence Strategy (MA + RSI)

Finally, I combined the long-term trend filter (200-day MA) with the short-term momentum oscillator (RSI). The logic requires both conditions to align: the asset must be in an established macro uptrend, but experiencing a short-term oversold dip. This strategy achieved 0.586 Sharpe ratio while limiting the drawdown to ~34% and maintaining a win ratio of 90%. Using a wider ATR of 3.5x prevented premature exits. While this did not beat cumulative gains for buy-and-hold, its trading efficiency and high win rates were compelling.

Evaluating Risk vs. Reward

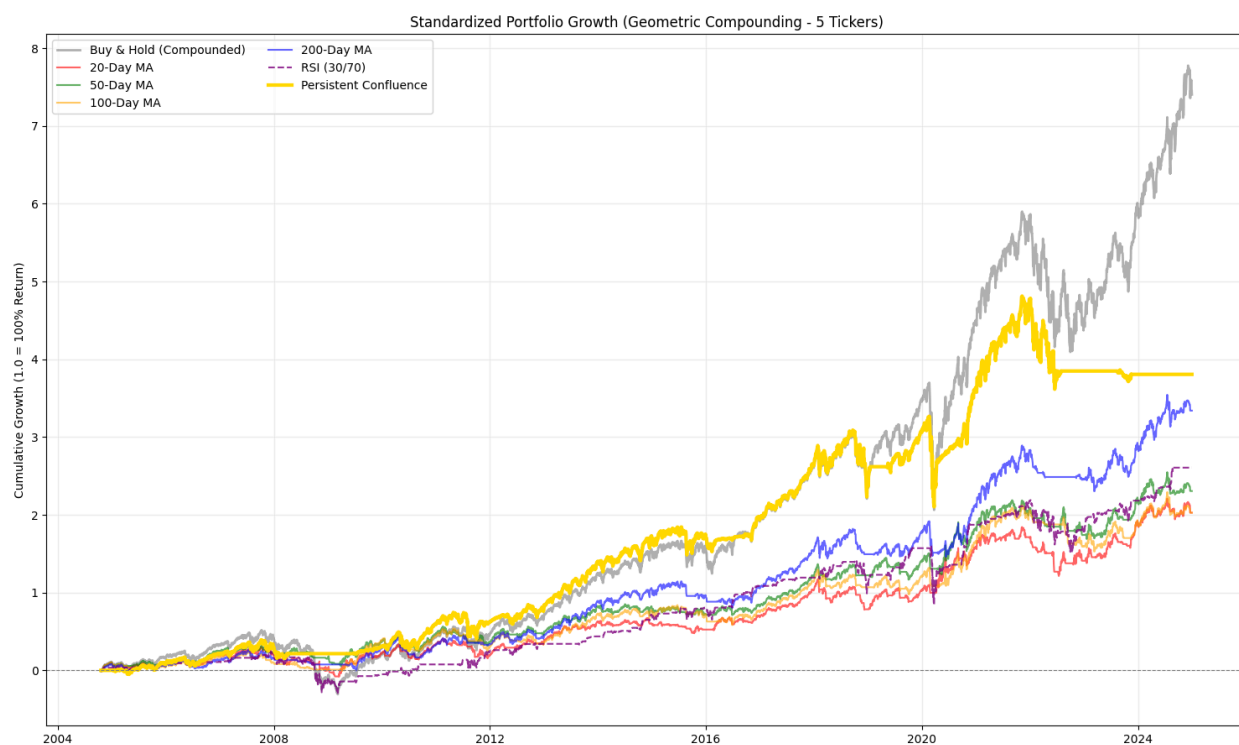
To figure out if these strategies were actually worth the effort over simply holding the indices, I calculated the maximum drawdown and evaluated the risk-adjusted performance using the Sharpe Ratio formula:

Sharpe Ratio = (Portfolio Return - Risk-Free Rate) / Standard Deviation of Portfolio Return

Aggregated Performance Findings (20-Year Portfolio Average)

Strategy	Total Return	Max Drawdown	Sharpe Ratio	Win Rate (%)	Exp Ret/Trade	Std Dev/Trade	Trades
200-Day MA	3.47	-0.242	0.627	52.92%	0.0280	0.0900	302
Buy & Hold	7.91	-0.549	0.619	100.0%	7.9099	0.0	5
Confluence (MA+RSI)	3.91	-0.344	0.586	89.33%	0.3816	0.3830	27
50-Day MA	2.40	-0.244	0.547	50.41%	0.0202	0.0746	312
100-Day MA	2.03	-0.261	0.504	48.62%	0.0219	0.0830	301

Strategy	Total Return	Max Drawdown	Sharpe Ratio	Win Rate (%)	Exp Ret/Trade	Std Dev/Trade	Trades
20-Day MA	2.19	-0.328	0.490	48.57%	0.0149	0.0678	421



Conclusion

My initial hypothesis asked if momentum could beat Buy & Hold.

1. **The Power of Compounding:** Bogle's passive indexing remains the undisputed leader in raw growth, crushing the algorithmic models over a 20-year window. However, it came with extreme volatility, forcing an investor to stomach a brutal 55% drawdown.
2. **Adaptive Risk Boosts Trend Following:** Replacing a static stop-loss with an adaptive ATR 4.0x multiplier on the 200-Day MA proved highly effective, boosting returns by over 50 percentage points compared to fixed stops while keeping the maximum drawdown contained to roughly 20%.
3. **Capital Preservation:** My custom Confluence strategy (MA + RSI) proved the most conservative. Initially, I found the high win rate particularly appealing because it created smoother and more psychologically consistent performance. However, I later realized that the win rate alone could be misleading if achieved at the cost of lower long-term returns or reduced exposure to major market trends.

Ultimately, building this Python model taught me that beating the long-term compounding of the market is incredibly difficult. However, algorithmic strategies provide immense value not by boosting pure returns, but by mathematically controlling risk and preserving capital during market stress.

My takeaways

This project changed how I think about investing and modeling. I initially expected increasingly sophisticated trading rules to consistently outperform passive investing. Instead, I found that additional complexity often introduced new tradeoffs, and that long-term compounding remained extremely difficult to beat.

Known Limitations & Areas for Future Research

While this backtest provides valuable insights into momentum trading, there are several structural limitations in this model:

- **Frictionless Trading Assumption:** The code currently assumes trades are executed perfectly without broker commissions, bid-ask spread slippage, or SEC fees. The high trading frequency of the shorter-window strategies would see their expected returns severely eroded by transaction costs.
- **Tax Drag:** The Bogle-style Buy & Hold strategy benefits heavily from deferred long-term capital gains. By contrast, the active strategies frequently trigger short-term capital gains taxes, which would drag down real-world portfolio performance.
- **Parameter Sensitivity & Overfitting:** Hyperparameters (like the 2% epsilon buffer and 4.0x ATR multiplier) were settled on after manual testing. There is a risk of "curve-fitting," meaning these specific parameters worked perfectly for this exact 20-year window, but might not be optimal for a future market regime.
- **Historical Macro Bias:** The 2004–2024 backtesting window encompasses a historic period of declining interest rates and quantitative easing that heavily favored a long-biased approach.
- **Limited Asset Universe:** The experiment was restricted to five highly liquid US ETFs. The findings do not necessarily apply to individual equities, which possess much higher idiosyncratic risk.
- **Long-Only Constraints:** Because the algorithm does not short-sell, the strategy can only play defense during bear markets (moving to cash) rather than actively profiting from downward momentum.

Technical Challenges & Iteration

- **Correcting Time-Series Volatility in Risk Metrics**

Early in my testing, the strategies reported unexpected Sharpe Ratios (in some cases, exceeding 10). Digging into the underlying mathematics, I discovered a fundamental flaw in my initial logic: I was calculating the ratio using a sparse list of individual trade returns rather than a continuous daily portfolio equity curve. Consequently, applying the $\sqrt{252}$ annualization factor to non-daily data severely skewed the volatility metric and completely ignored the "cash drag"—the extended periods the algorithm spent out of the market earning 0%. By implementing a function to create a continuous daily time-series, the Sharpe Ratio now correctly accounts for true daily volatility and the opportunity cost of time spent in cash.

- **Aligning Geometric and Arithmetic Compounding**

During the evaluation phase, I noticed a significant discrepancy between my visualizations and data tables: the growth plots displayed approximately 200% growth, while the performance tables showed roughly 390%. This occurred because the plotting logic was calculating an arithmetic sum, whereas the data tables correctly utilized geometric compounding. By updating the visualization logic to use a cumulative product (`cumprod`) on the daily returns, I ensured both the charts and tables consistently and accurately reflected the geometric compounding of market returns.

- **Mitigating Stop-Loss "Churn" via Tightness Analysis**

When first implementing an Average True Range (ATR) stop-loss, the strategy experienced a 300% increase in trade "churn" (frequent exits and re-entries), which became a significant performance drag. To diagnose this inefficiency, I engineered a "Stop Tightness Analysis," which revealed that the initial ATR-based stops were mathematically too tight (averaging around a 2.6% drop) compared to a broader 5% target. This quantitative analysis allowed me to optimize the framework, prioritizing a 5% fixed trailing stop in these specific instances to ensure better long-term trend retention and reduce over-trading.

- **Ensuring Data Alignment and Uniform Time Horizons**

A critical structural flaw emerged when comparing strategies with different baseline requirements. Initially, the passive Buy & Hold baseline began evaluating on Day 1, while the 200-day Moving Average strategy required 200 days of historical data before it could generate its first mathematical signal. This resulted in a misaligned, unfair comparison. To resolve this, I standardized the core backtesting function to enforce a mandatory 200-day "warm-up" period across the board. This guaranteed that all strategies were evaluated over the exact same historical time window, ensuring strict data integrity.